HEIG-VD HAUTE ÉCOLE D'INGÉNIERIE ET DE GESTION DU CANTON DE VAUD

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Formal verification
Coverage
CERN training - November 2026

Context

We are verifying a blackbox sequencer who has the responsibility of generating certain sequences. Its entity is the following :

```
entity blackbox_sequencer is
    generic (
        ERRNO : integer := 0
    );
    port (
        clk : in  std_logic;
        a   : out std_logic;
        b   : out std_logic;
        c   : out std_logic;
        d   : out std_logic;
        e   : out std_logic;
        f   : out std_logic
    );
end blackbox_sequencer;
```

1. We expect to observe the three-cycle pattern in which `d` is high, then `e` is high on the next cycle, and then `d` is high again. This pattern should occur twice. How could we check that ?
2. If `f` is high on two consecutive clock cycles, then on the following cycle `e` should be high. Can you check this requirement ?
3. Whenever `a` is high, check whether `c` is high on that cycle and the following cycle. If so, then on the next three cycles `b` should be high, low, and high, in that order.

   Let's start by writing an assertion that just checks that with a double implication, as in this previous sentence.

   Sounds good ?

   Let's talk about what we can do.

Running the coverage and assertion checks

The solution provides four configurations, `ERRNO=0` through `ERRNO=3`.
- `ERRNO=0` is the reference behavior
- `ERRNO=1` changes only the sequence of `f` compared with `ERRNO=0`
- `ERRNO=2` changes only `b` compared with `ERRNO=0`
- `ERRNO=3` changes `a` and `b` compared with `ERRNO=0`

The `run.sby` script defines `coverX` and `bmcX` tasks for each value X.
Start with the reference configuration, `ERRNO=0`, check the assertions and check the coverage goals.

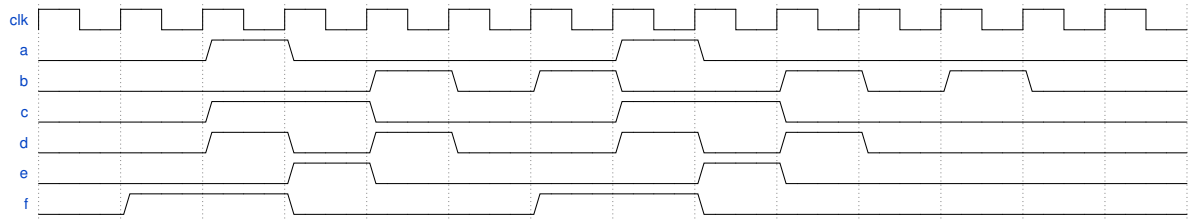
```
sby --yosys "yosys -m ghdl" -f run.sby bmc0
sby --yosys "yosys -m ghdl" -f run.sby cover0
```

Afterwards, run the matching tasks for `ERRNO=1`, `ERRNO=2`, and `ERRNO=3`. Verify the output and check the expected failures. An "expected failure" is a deliberate negative test configured in the script, not a successful proof of the assertions or coverage goals.

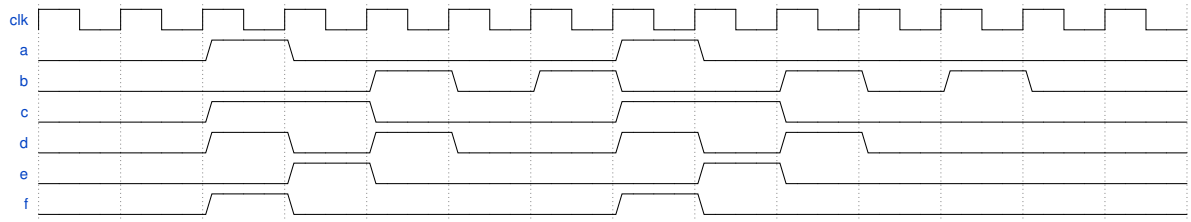
The expected-failure tasks in the script are `cover1`, `cover3` and `bmc2`. All other tasks are configured as ordinary passes.

Reference chronograms

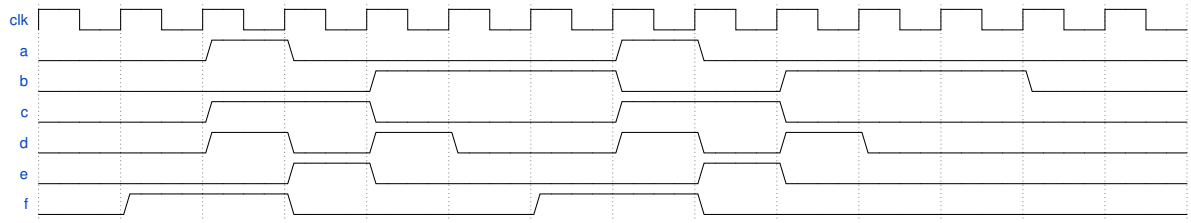
Chronogram for `ERRNO=0`



Chronogram for `ERRNO=1`



Chronogram for `ERRNO=2`



Chronogram for `ERRNO=3`

